

元智大學
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碩士論文計畫書
MASTER'S THESIS PROPOSAL

隱形帳簿：量化印尼平台經濟中的隱形楔子
The Invisible Ledger: Quantifying the Invisible Wedge in Indonesia's Platform Economy

研究生 / Student: 王新福 Christopher Ongko
指導教授 / Advisor: 孔德蓉 De-Rong Kong

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Abstract

The economic measurement of platform-based commerce rests on records that do not describe the same thing. Platform revenue, e-commerce statistics and payment data are commonly used as interchangeable indicators of digital activity, although each captures a different side of the boundary between the value a platform processes and the revenue it records. This paper measures that boundary for Indonesia, where I term it the invisible wedge, using issuer disclosures for 2020-2025. In FY2023, three documented platform cases processed transaction value 13.7 times their recognised revenue, a wedge of US\$40.07 billion or 2.9 percent of GDP, and the wedge stays above US\$20 billion when any single platform is excluded. Within platform series, revenue grew faster than transaction value in 10 of 12 annual transitions, with a median divergence of 42 percentage points. Official statistics place 98.5 percent of Indonesia's 2023-2024 e-commerce growth outside the marketplace channel that platform accounts record, while digital payment value grew up to eleven times as fast as e-commerce value. Each of these records misstates both the size of the digital economy and where it is growing.

1. Introduction

Southeast Asia's digital economy grew from US\$100 billion in gross transaction value in 2020 to US\$263 billion in 2024 (Google, Temasek and Bain 2020, 2024; reported there as gross merchandise value). The numbers used to describe that growth come from three kinds of record: platform accounts, which report what platforms earn; e-commerce and marketplace statistics, which report sales through particular channels; and payment statistics, which report the value moving through digital payment systems. They are commonly used as interchangeable measures of the same activity. This proposal argues that they are not interchangeable, and that treating them as such produces systematically wrong measurement of platform economies. In Indonesia, platform revenue understates the level of the commerce platforms carry by more than an order of magnitude and overstates its growth, marketplace statistics miss almost all of the recent expansion, and payment data grow many times faster than the commerce they settle.

Suppose a platform processes 100 units of transaction value and recognises 10 as revenue. The other 90 units are recorded in the platform's systems as merchant receipts, driver payouts, inventory costs and other pass-through payments, and never appear as platform revenue. An analyst sizing the platform from its accounts sees 10; an analyst sizing it from its transaction data sees 100. I call the difference the invisible wedge.

For FY2023, three documented Indonesian platform cases processed US\$43.23 billion of transaction value against US\$3.16 billion of recognised revenue, leaving a wedge of US\$40.07 billion. The wedge alone is equivalent to 2.9 percent of Indonesia's US\$1.371 trillion GDP. It remains above US\$20 billion when any one platform is removed, and it moves by less than half a billion dollars across the plausible range of its key monetisation assumption, as Section 5.1 shows.

The wedge is invisible in a specific sense. Platforms hold detailed, seller-linked records of the full transaction value, but only the revenue share reaches their financial statements, and the transaction records themselves do not routinely enter third-party income reporting. Indonesia's Regulation PMK 37/2025 is the first attempt to route those records to the tax authority, and Section 5.3 shows how small a part of the activity its design can reach.

1.1 Research question and objectives

One question drives the proposal:

How large is the gap between the commerce Indonesian platforms carry and the revenue their accounts record, how does it move over time, and what does it imply for how the digital economy is measured?

The proposal pursues this through three objectives. It measures the wedge across FY2020-FY2025 for five Indonesian platform series, keeping observations separated by how directly they map to

Indonesia. It explains the wedge's movements using revenue components the platforms themselves disclose. And it sets platform-based measures against national e-commerce statistics and payment data to establish how far each departs from the others. Evidence from 48 issuer-years of platforms outside Indonesia tests whether the same boundary appears in other markets and business models (Section 5.3).

1.2 Contributions

First, a longitudinal measure of the wedge. The evidence extends from a single-year snapshot to seventeen platform-year observations over FY2020-FY2025, and shows that the ratio of transaction value to revenue changes substantially within the same platform from year to year.

Second, a mechanism. Material movements in the wedge are traced to disclosed monetisation, customer incentives, gross-versus-net recognition and business perimeter. In the cleanest case, Tokopedia's net revenue rose 53.2 percent in a year in which its transaction value fell 8.9 percent, mostly through lower customer incentives.

Third, a demonstration that the standard measures of digital commerce are not interchangeable. Combining issuer evidence with national statistics from BPS-Statistics Indonesia (Badan Pusat Statistik), Bank Indonesia payment data and Regulation PMK 37/2025 (Peraturan Menteri Keuangan), administered by the Directorate General of Taxes (DJP), the thesis shows that platform revenue, marketplace statistics and payment data each misstate Indonesian digital commerce in a different direction, and that the new marketplace reporting regime inherits the limits of the channel it targets.

1.3 Positioning the contribution

The contribution is a finance measurement problem before it is a tax-policy one. Investors, statistical agencies and tax authorities each draw conclusions about the scale, growth and monetisation of platform businesses from whichever record they have to hand, and those conclusions differ because the records differ. De Franco, Kothari and Verdi (2011) show that users draw better inferences when accounting bases are comparable, and Berg et al. (2020) show that intermediary-held digital traces carry information conventional records lack. This proposal applies both results to platform economies and shows that the choice of record, usually treated as a question of data availability, changes the answer.

2. Literature Review and Research Gap

2.1 Platform economics and revenue recognition

Multisided-platform theory explains why a platform's revenue need not track the transaction value it coordinates (Rochet and Tirole 2003; Caillaud and Jullien 2003; Parker and Van Alstyne 2005; Armstrong 2006; Hagiu and Wright 2015; Evans and Schmalensee 2016). Accounting determines how much of that value becomes revenue: principal-agent treatment under IFRS 15 (International Accounting Standards Board 2014), customer incentives, service mix, acquisitions and reporting perimeter all move revenue independently of the underlying commerce. De Franco, Kothari and Verdi (2011) show why comparable accounting bases matter when users interpret such movements. What this literature has not done is measure how large and how unstable the resulting gap is within a single economy over time.

2.2 Informality, digital records, and third-party information

Platform participants occupy an unusual position: their transactions are priced, recorded and settled through a formal intermediary while the participants themselves may be self-employed, unregistered or below filing thresholds. That places the invisible wedge outside the shadow-economy framework of the informality literature (La Porta and Shleifer 2014; Ulyssea 2018; Medina and Schneider 2019), because the activity is fully recorded; it is simply recorded somewhere other than where it is measured. Such digital records carry economic information even where conventional records are thin (Berg et al. 2020), and platform-mediated work leaves

observable financial and administrative traces (Barrios, Hochberg and Yi 2022; Denes, Lagaras and Tsoutsoura 2025).

A central public-finance result is that third-party information materially changes compliance and enforcement (Kleven et al. 2011; Pomeranz 2015; Naritomi 2019; Kleven, Kreiner and Saez 2016; Slemrod 2019). The OECD Model Rules and the European Union's DAC7 regime put that result into practice for platforms, and Indonesia's PMK 37/2025 applies the same logic through seller identification, transaction-linked reporting and marketplace withholding.

2.3 Measuring the digital economy

Digital-economy measurement frameworks recognise the same boundary from the national-accounting side. Official guidance separates the buyer-seller transaction from the intermediation service the platform provides, and treats only the service as the platform's output (Ahmad and Schreyer 2016; International Monetary Fund 2018; OECD 2023; United Nations et al. 2025). The frameworks are clear in principle, yet the indicators most often quoted for the digital economy, gross merchandise value, platform revenue and payment volume, each sit on a different side of that line. This proposal takes the frameworks' distinction seriously and measures what happens when those indicators are set against one another.

2.4 Research gap

The literature explains why transaction value can exceed platform revenue and why accounting moves the two apart. It has not measured the gap longitudinally for one economy, traced its movements to disclosed components, or tested how far the standard indicators of digital commerce depart from one another, and this proposal does all three for Indonesia.

3. Theoretical Framework and Key Variables

Let V denote transaction value and R platform-recognized revenue for a matched platform scope and period. The core measures are:

$$W = V - R \quad E = (V - R) / R \quad D = g(V) - g(R)$$

W is the absolute invisible wedge; E , the Ecosystem Ratio, is constructed for this study and measures the wedge per unit of platform revenue; and D is the annual growth divergence between transaction value and revenue. A negative D means platform revenue is growing faster than the commerce behind it.

Gross transaction value (GTV) is the proposal's common label for the value processed through a platform. Issuer terminology differs: Sea Limited and Blibli use GMV, Grab uses GMV for deliveries and TPV for financial services, and GoTo uses GTV. The source labels and business perimeters are retained in the empirical files, and the main longitudinal comparisons are made within named series.

3.1 Hypotheses

The framework yields four hypotheses, set out in Table 1 with the evidence already assembled for each. H_4 is the central claim of the thesis, and H_1 to H_3 establish the mechanisms behind it. Whether platform records become administratively usable once PMK 37/2025 takes effect on 1 November 2026 depends on post-implementation evidence; Section 5.3 addresses it through the design of the regime.

Table 1. Hypotheses and current evidence.

Hypothesis	Status in the current evidence
H1 Platform revenue is not a stable proxy for platform commerce: the ratio of transaction value to revenue moves materially within the same platform over time.	Supported. Revenue outgrows transaction value in 10 of 12 within-series transitions; median divergence 42.02 pp. Survives log growth (36.64), removal of the largest transition (29.34 pp) and exclusion of any single series.
H2 Indonesian e-commerce growth is driven by the entry of new businesses rather than by higher value per business.	Supported. The business-count term accounts for 90.29% of the 2023-2024 increase in e-commerce value, and 70.95% under the conflicting BPS business count.
H3 Marketplace participation is associated with financial recordkeeping.	Supported in BPS business-level evidence. The within-province change is weaker ($r = 0.309$, $p = 0.066$) and is tested further in the thesis.
H4 Platform revenue, marketplace statistics and payment data are not interchangeable measures of digital commerce, and substituting one for another misstates its scale and location.	Supported. FY2023 transaction value is 13.7 times revenue; 98.46% of 2023-2024 e-commerce growth falls outside the marketplace channel; payment value grew 1.8-11 times as fast as e-commerce value.

4. Data and Methodology

4.1 Admission rules and evidence tiers

A platform-period is eligible only when transaction value and revenue cover the same period; geography and business scope can be evaluated; units and definitions are known; derived inputs trace to source; and structural breaks or revised reporting bases are flagged. Repeated publication vintages are not counted as independent observations. Final admission to the main sample is decided at the proposal examination (Section 7).

Table 2. Evidence tiers for Indonesian platform observations. Tiers are reported separately according to how directly each series maps to Indonesia.

Evidence tier	Series	Levels	Transitions
Direct Indonesia-aligned	Tokopedia e-commerce	2	1
Direct, scope-pending	Blibli 3P; Bukalapak Group	9	7
Conditional reconstruction	Grab; Shopee Indonesia	6	4
All tiers (diagnostic)	Five series	17	12

Table 2 shows the trade-off the design manages: only two of the seventeen levels come from a directly Indonesia-aligned segment, while the larger tiers carry more observations under wider scope. Tokopedia is the strongest Indonesia-aligned series; Blibli includes online travel; Bukalapak reports at Group level with overseas operations; and Grab and Shopee each require a derived or externally estimated country component. Tokopedia FY2021 and Bukalapak FY2024 are excluded because their transaction and revenue periods do not match.

Three counts are intentionally different. The all-tier inventory contains 17 retained levels; the executed direct tiers contain 11 levels (2 Indonesia-aligned and 9 scope-pending); and the broader direct-candidate sensitivity contains 13 periods, including a Blibli FY2020 prospectus observation, yielding 9 annual transitions. These counts answer different construction questions and are reported separately.

4.2 Comparability strategy

The main longitudinal comparisons are made within named series. A within-series transition holds the issuer, business perimeter and disclosure convention as constant as the filings allow, so a change in the ratio of transaction value to revenue reflects a change in that platform's economics and accounting.

Cross-platform levels serve a different purpose. The FY2023 cross-section establishes scale, and the 48 external issuer-years test whether the boundary appears outside Indonesia. Keeping those roles distinct is what allows heterogeneous disclosures to be used together.

4.3 Conditional country evidence

Grab discloses Indonesia revenue but not Indonesia transaction value. Its conditional country transaction value is therefore derived as:

$$\text{Indonesia GTV} = \text{Indonesia revenue} \times \text{Group GTV} / \text{Group revenue}$$

The construction applies the Group monetisation rate to Indonesia. Shopee's Indonesian transaction value comes from Momentum Works' recurring Southeast Asian e-commerce estimates, since Sea Limited discloses no country figure, and is labelled as third-party wherever it is used.

5. Preliminary Evidence and Feasibility

5.1 FY2023 scale

Table 3. Indonesia-focused FY2023 cross-section, US\$ billions. V is transaction value, R platform revenue, W the wedge and E the Ecosystem Ratio; the final row sums the three cases, and E is computed from unrounded inputs.

Case	V	R	W	E	Evidence class
Grab Indonesia	5.381	0.605	4.776	7.895x	Derived V
Tokopedia	16.331	0.405	15.926	39.296x	Direct pair
Shopee Indonesia	21.520	2.152	19.368	9.000x	Derived V and R
Selected platforms	43.233	3.162	40.070	12.671x	Sum of three cases

Table 3 reports the FY2023 cross-section. Across the three cases, transaction value is 13.7 times recognised revenue. Tokopedia, the one direct pair, contributes 39.7 percent of the wedge, Shopee 48.3 percent and Grab 11.9 percent. Shopee's Indonesian revenue is estimated by applying Sea Limited's disclosed Group monetisation rate of 10.0 percent to an external market estimate, which fixes its Ecosystem Ratio at 9.000; varying that rate between 9 and 11 percent moves the combined wedge only between US\$40.29 billion and US\$39.86 billion, because a higher assumed rate raises estimated revenue and lowers the residual almost equally. Excluding any single platform leaves a wedge of between US\$20.70 billion and US\$35.29 billion, and one-at-a-time variation of every derived input keeps it between US\$37.65 billion and US\$42.49 billion. Tokopedia alone, which rests on no reconstruction, carries a wedge of US\$15.93 billion.

5.2 Longitudinal movement and mechanism

Table 4. Within-series annual divergence under two admission rules. Counts differ because the admission rules differ.

	All-tier inventory	Direct-candidate sensitivity
Transitions	12	9
Revenue grows faster	10	6
Transaction grows faster	2	3
Opposite-sign transitions	2	3
Median absolute divergence	42.02 pp	42.94 pp

Table 4 compares the two admission rules, and both lead to the same conclusion. Under the all-tier inventory, revenue grew faster than transaction value in 10 of 12 annual transitions, with a median absolute divergence of 42.02 percentage points; under the direct-candidate rule, revenue grew faster in 6 of 9 and the median is 42.94 points. Platform revenue growth is therefore a poor guide to the growth of the commerce behind it, and in most years it overstates that growth. Under the direct-candidate rule the result survives the obvious tests: measured in log changes the median gap is 36.64 points, removing the single largest transition leaves 29.34 points, and every construction that excludes one platform series retains at least one sign reversal. Tokopedia FY2022-FY2023 shows the mechanism: transaction value fell 8.90 percent while third-party net segment revenue rose 53.20 percent, and 60.56 percent of that revenue gain came from lower customer incentives, with 39.44 percent from higher gross revenue.

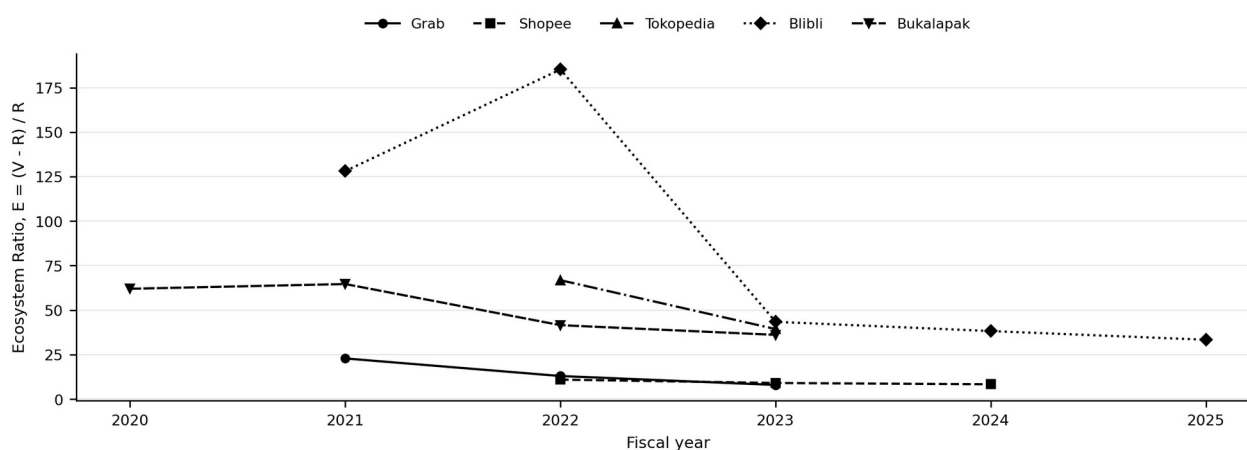


Figure 1. Ecosystem Ratio across the extended candidate series, FY2020-FY2025. Each line follows one disclosed or reconstructed series.

Figure 1 shows the pattern behind Table 4. The wedge is present in every series in every year, at ratios from about 8 to over 180, and in every series it is lower at the end of the period than at the start, as platforms have raised their monetisation. Measured through platform revenue, these businesses appear to be growing faster than the commerce they carry, which is the distortion H1 describes.

5.3 Wider Indonesian activity and reporting

BPS reports national e-commerce transaction value rising 17.08 percent from 2023 to 2024. The marketplace component, the channel that platform accounts describe, rose 1.45 percent while the non-marketplace component rose 20.57 percent, so 98.46 percent of the increase took place outside the marketplace channel. The marketplace share of e-commerce value fell from 18.2 to 15.8 percent in a single year, so measurement anchored on marketplace platforms misses almost all of the recent growth. Growth in the number of e-commerce businesses accounts for 90.29 percent of the national increase, with value per business contributing the remainder (H2).

PMK 37/2025 is the institutional response. The verified implementation sequence records marketplace designation on 1 July 2026, collection effective 1 August, postponement through 31 October, and scheduled implementation on 1 November 2026. A platform record becomes usable for tax administration only once a seller identity is attached, the record is transmitted to the DJP and it is matched to a taxpayer. The regime designates marketplace operators, and the marketplace channel carried 15.8 percent of 2024 e-commerce value and 1.5 percent of its 2023-2024 growth. However well it is implemented, its coverage is bounded by the channel it targets.

Outside Indonesia, 48 matched issuer-years across eight platform businesses, including eBay, Etsy, Shopify, Mercado Libre and Sea, show the same boundary under other business models. Eleven of 40 annual transitions fail a clean-scope test because of acquisitions or perimeter changes, which indicates that the comparability problem the evidence tiers address is a feature of platform disclosure generally.

5.4 Payment traces and administrative records

Bank Indonesia's payment statistics provide a third record of the same activity. Between 2023 and 2024, BPS e-commerce value grew 17.08 percent, while electronic-money shopping value grew 30.47 percent, mobile-banking payment and purchase value 82.84 percent and QRIS transaction value 186.98 percent. Payments and sales are different objects, and the payment series grew between 1.8 and 11 times as fast as the commerce they help settle, so a digital-economy indicator built on payment value would substantially overstate commerce growth over this period.

5.5 What the preliminary evidence establishes

The preliminary evidence supports the central claim at each step. Platform revenue understates the level of platform commerce by a factor of 13.7 and, in most years, overstates its growth.

Marketplace statistics miss 98.5 percent of the recent expansion. Payment data grow up to eleven times as fast as the commerce behind them. And the one disclosed decomposition traces the revenue-transaction gap to identifiable commercial choices. The thesis will extend and test these results under the sample boundary approved at examination.

6. Why the Problem Matters

The consequences follow directly. An investor valuing platforms on revenue growth, a statistical agency sizing e-commerce from marketplace data and a policymaker reading payment growth as commerce growth are each working from a record that departs sharply from the activity they intend to measure, and in a different direction. The same transformation looks like rapidly monetising platforms, a stagnant marketplace sector or an explosion of digital commerce depending on which record is used. Indonesia's new reporting regime inherits the problem, because it is built around the marketplace channel. The thesis identifies proxy substitution between these records as the underlying error, and measures its size.

7. Robustness and Proposal Decisions

Table 5. Claims, evidence and robustness tests.

Claim	Evidence and robustness test
Platform revenue understates platform commerce by more than an order of magnitude.	FY2023 transaction value is 13.7 times revenue (wedge US\$40.07bn). Holds across Shopee monetisation of 9-11% (US\$39.86-40.29bn), leave-one-platform-out (at least US\$20.70bn), one-at-a-time input variation (US\$37.65-42.49bn), and on the Tokopedia direct pair alone (US\$15.93bn).
Platform revenue growth misstates commerce growth.	Revenue outgrows transaction value in 10 of 12 transitions; median divergence 42.02 pp. Holds in log changes (36.64), without the largest transition (29.34 pp), under the direct-candidate rule (42.94 pp) and in every leave-one-series-out construction.
Marketplace statistics miss where e-commerce grows.	98.46% of 2023-2024 growth in e-commerce value falls outside the marketplace component; marketplace share 18.2% to 15.8%. Holds on both the published amount and the rounded-share reconstruction (98.49%).
Payment data overstate commerce growth.	Electronic-money, mobile-banking and QRIS payment value grew 1.8 to 11 times as fast as e-commerce value, 2023-2024.
Questions the thesis tests further.	Causal effects of incentive choices; business-level validation of the BPS recordkeeping association (province $r = 0.309$, $p = 0.066$); operational performance of PMK 37/2025 after 1 November 2026.

Table 5 sets each claim against its evidence and the robustness tests it survives. The sample is small and the evidence descriptive, so the thesis reports tests rather than population estimates. What it establishes is that the standard records of digital commerce diverge by magnitudes too large for measurement or policy to ignore.

Two sample decisions are put to the committee: whether Blibli 3P Retail and Bukalapak Group enter the final longitudinal core, and whether Grab and Shopee remain sensitivity evidence. The central claim holds under either outcome: the divergence result holds under both admission rules in Table 4, and the level result holds on the Tokopedia pair alone.

8. Next Step

After the proposal examination the approved sample boundary will be frozen and the requested checks completed during October-November 2026, with the full manuscript rebuilt around that boundary before the final defence. Robustness, ASEAN/payment extensions, detailed source reconciliation, and the complete verification ledger remain available in the repository and advisor data package. Table 6 sets out that schedule; the empirical backend is already assembled, so what remains is sample admission, requested analysis, and manuscript construction.

Table 6. Planned work to the final defence.

Period	Planned work
September 2026	Advisor review of the empirical package; proposal oral examination; freeze the sample-admission rule and the checks the committee requests.
October-November 2026	Complete requested reconciliations and source concordance; extend sensitivity to the approved tier boundary; freeze thesis tables and figures.
Before the final defence	Rebuild the manuscript under the approved sample; retain supporting payment, ASEAN and global modules only where they advance the final argument.

The proposal establishes that the records used to measure Indonesia's platform economy diverge far more than their interchangeable use assumes. Platform revenue understates the level of platform commerce by more than an order of magnitude and misstates its growth, marketplace statistics miss almost all of the recent expansion, and payment data outrun commerce by up to eleven times. What remains is to fix the sample boundary, complete the checks the committee requires and build the thesis on that decision; the data, the evidence tiers and the reproducible backend are already in place.

Appendix A - data sources by evidence class

Table A1. Source lineage by evidence class.

Tier	Series	Source of V	Source of R
Direct Indonesia-aligned	Tokopedia e-commerce	GoTo annual report, e-commerce segment metrics	GoTo annual report, segment note
Direct, scope-pending	Blibli 3P Retail	Global Digital Niaga prospectus and results releases	Same issuer filings
Direct, scope-pending	Bukalapak Group	Bukalapak annual and sustainability reports	Same issuer filings
Conditional reconstruction	Grab Indonesia	Derived from disclosed Indonesia revenue and Group monetization rate (Form 20-F)	Grab Form 20-F
Conditional reconstruction	Shopee Indonesia	Momentum Works Southeast Asia e-commerce estimate (third-party)	Derived from Sea Limited Form 20-F
National statistics	BPS-Statistics Indonesia	E-Commerce Statistics 2023, 2024; BPS directorate presentation	-
Regulatory	PMK 37/2025; DJP	Ministry of Finance of the Republic of Indonesia	-

Every figure used in this proposal traces to a named file and locator in the accompanying data package.

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